ASSIGNMENT – VII

TITLE :
Configure a multi-node Hadoop cluster and process large datasets for a business intelligence solution.

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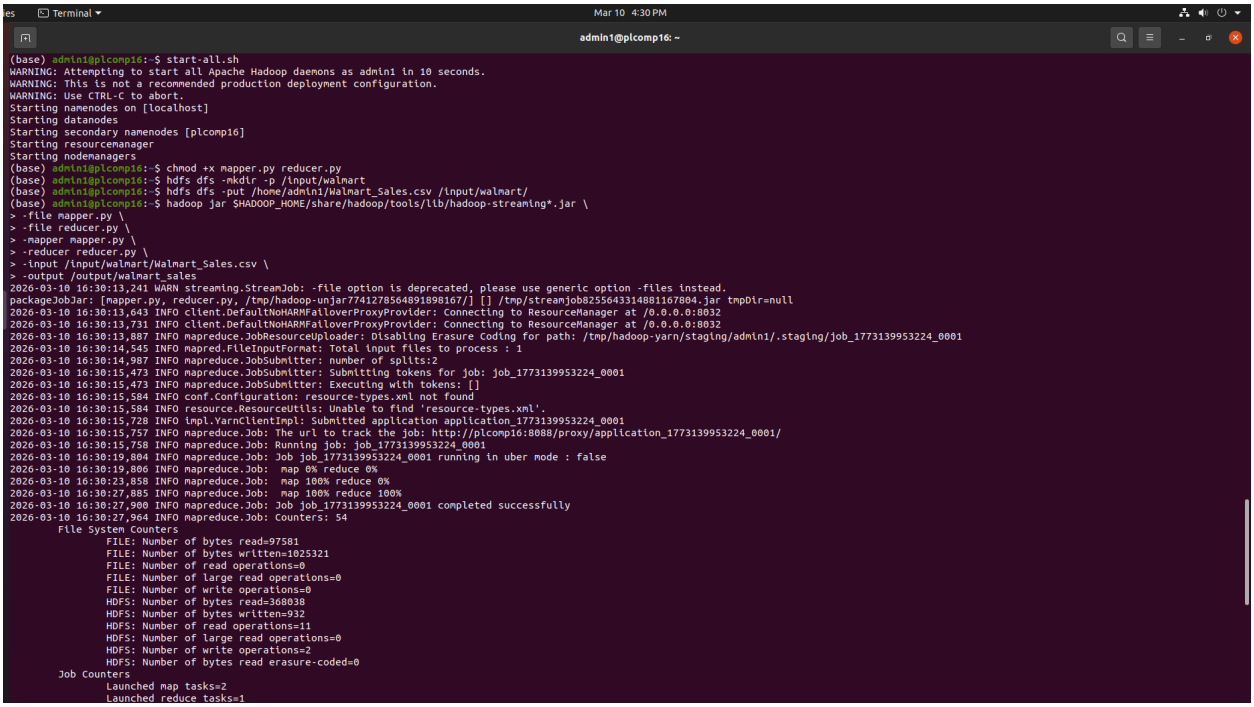
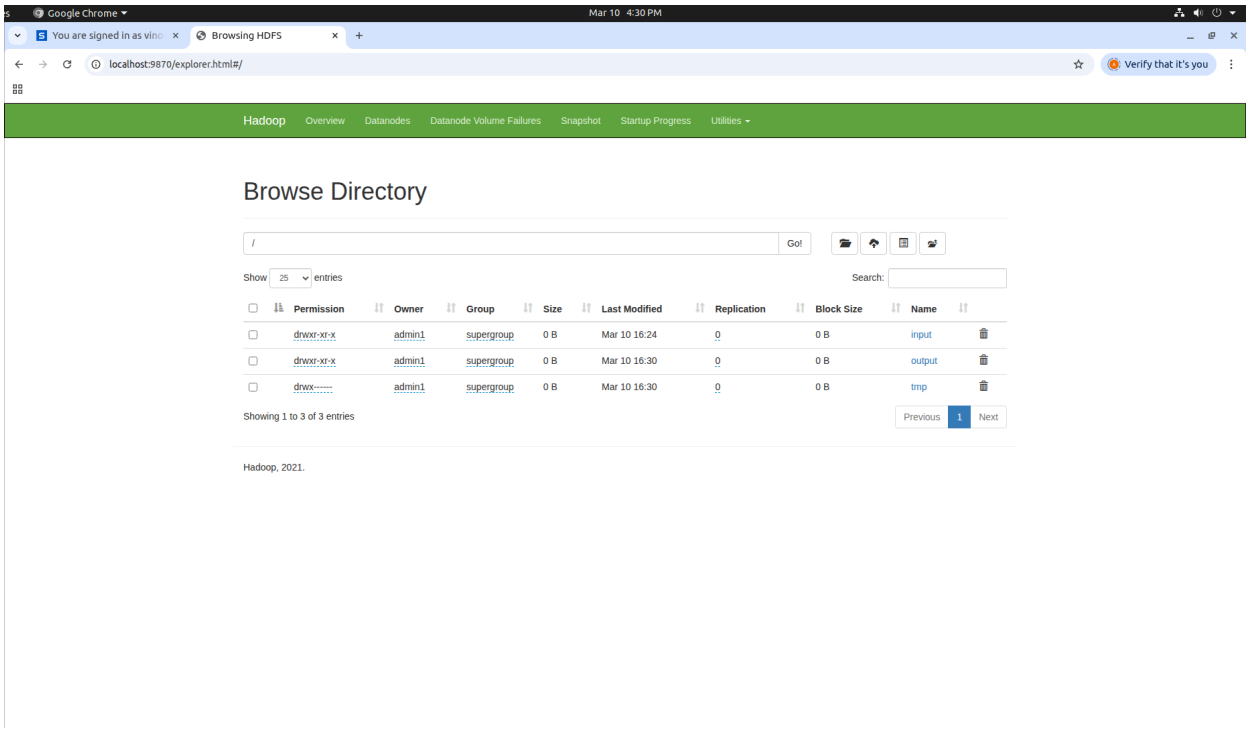
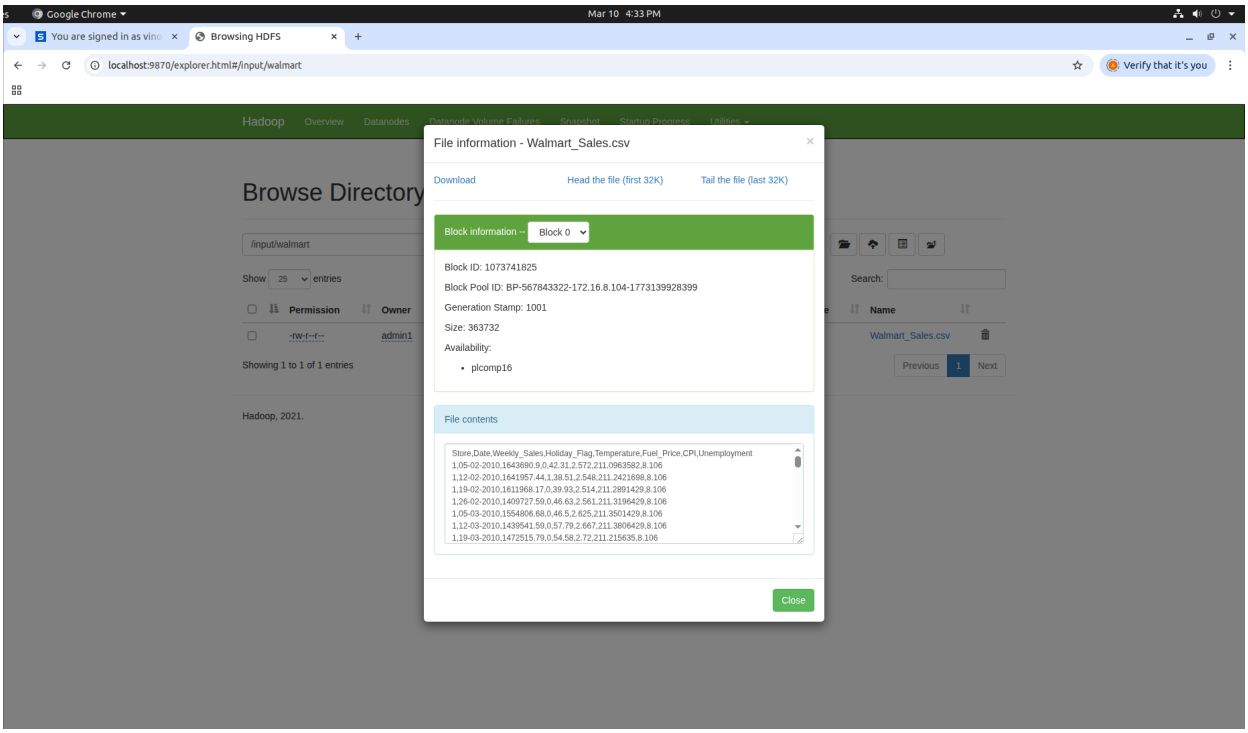
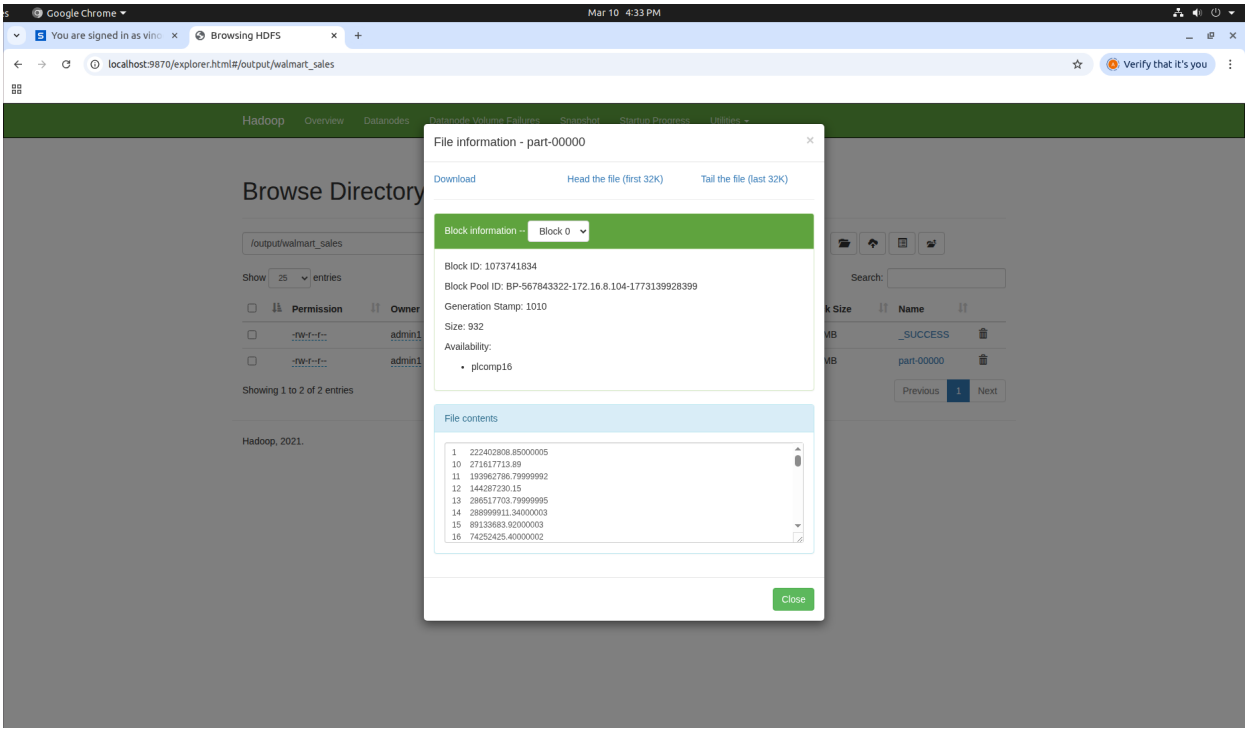


```
(base) admin1@plcomp16:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as admin1 in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [plcomp16]
Starting resourcemanager
Starting nodemanagers
(base) admin1@plcomp16:~$ chmod +x mapper.py reducer.py
(base) admin1@plcomp16:~$ hdfs dfs -mkdir -p /input/walmart
(base) admin1@plcomp16:~$ hdfs dfs -put /home/admin1/Walmart_Sales.csv /input/walmart/
(base) admin1@plcomp16:~$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming*.jar \
> -file mapper.py \
> -file reducer.py \
> -mapper mapper.py \
> -reducer reducer.py \
> -input /input/walmart/Walmart_Sales.csv \
> -output /output/walmart_sales
2026-03-10 16:30:13,241 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper.py, reducer.py, /tmp/hadoop-unjar7741278564891898167/] [] /tmp/streamjob8255643314881167804.jar tmpDir=null
2026-03-10 16:30:13,643 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-10 16:30:13,731 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-10 16:30:13,887 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/admin1/.staging/job_1773139953224_0001
2026-03-10 16:30:14,545 INFO mapred.FileInputFormat: Total input files to process : 1
2026-03-10 16:30:14,987 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-10 16:30:15,473 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1773139953224_0001
2026-03-10 16:30:15,473 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-10 16:30:15,584 INFO conf.Configuration: resource-types.xml not found
2026-03-10 16:30:15,584 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-10 16:30:15,728 INFO impl.YarnClientImpl: Submitted application application_1773139953224_0001
2026-03-10 16:30:15,757 INFO mapreduce.Job: The url to track the job: http://plcomp16:8088/proxy/application_1773139953224_0001/
2026-03-10 16:30:15,758 INFO mapreduce.Job: Running job: job_1773139953224_0001
2026-03-10 16:30:19,804 INFO mapreduce.Job: Job job_1773139953224_0001 running in uber mode : false
2026-03-10 16:30:19,806 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-10 16:30:23,858 INFO mapreduce.Job:  map 100% reduce 0%
2026-03-10 16:30:27,885 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-10 16:30:27,900 INFO mapreduce.Job: Job job_1773139953224_0001 completed successfully
2026-03-10 16:30:27,964 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=97581
                FILE: Number of bytes written=1025321
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=368038
                HDFS: Number of bytes written=932
                HDFS: Number of read operations=11
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
```




Google Chrome
Mar 10 4:33 PM
You are signed in as vinc
Browsing HDFS
localhost:9870/explorer.html#/input/walmart
Verify that it's you
Hadoop
Overview
Datanodes
Browse Directory
/input/walmart
Show 25 entries
Search:
Permission
Owner
Name
-rw-r--r--
admin1
Walmart_Sales.csv
Showing 1 to 1 of 1 entries
Previous
1
Next
Hadoop, 2021.
File information - Walmart_Sales.csv
Download
Head the file (first 32K)
Tail the file (last 32K)
Block information -- Block 0
Block ID: 1073741825
Block Pool ID: BP-567843322-172.16.8.104-1773139928399
Generation Stamp: 1001
Size: 363732
Availability:
plcomp16
File contents
Store,Date,Weekly_Sales,Holiday_Flag,Temperature,Fuel_Price,CPI,Unemployment
1,05-02-2010,1643690.9,0,42.31,2.572,211.0963582,8.106
1,12-02-2010,1641957.44,1,38.51,2.548,211.2421698,8.106
1,19-02-2010,1611968.17,0,39.93,2.514,211.2891429,8.106
1,26-02-2010,1409727.59,0,46.63,2.561,211.3196429,8.106
1,05-03-2010,1554806.68,0,46.5,2.625,211.3501429,8.106
1,12-03-2010,1439541.59,0,57.79,2.667,211.3806429,8.106
1,19-03-2010,1472515.79,0,54.58,2.72,211.215635,8.106
Close


Google Chrome
Mar 10 4:33 PM
You are signed in as vinc
Browsing HDFS
localhost:9870/explorer.html#/output/walmart_sales
Verify that it's you
Hadoop
Overview
Datanodes
Browse Directory
/output/walmart_sales
Show 25 entries
Search:
Permission
Owner
Name
-rw-r--r--
admin1
_SUCCESS
part-00000
Showing 1 to 2 of 2 entries
Previous
1
Next
Hadoop, 2021.
File information - part-00000
Download
Head the file (first 32K)
Tail the file (last 32K)
Block information -- Block 0
Block ID: 1073741834
Block Pool ID: BP-567843322-172.16.8.104-1773139928399
Generation Stamp: 1010
Size: 932
Availability:
plcomp16
File contents
1 222402808.85000005
10 271617713.89
11 193962786.79999992
12 144287230.15
13 286517703.79999995
14 288999911.34000003
15 89133683.92000003
16 74252425.40000002
Close